

Fractions, Decimals & Percentages

PAPER D — HARD

Name _____ Date _____ Score _____

Before you start

- No calculator. Some of these have a short route and a long route — the short route is usually a single multiplier.
- If the amount you are given is described as being **after** a change (“after a discount”, “including tax”, “20% more than”), you are dividing, not multiplying.
- Successive percentage changes **multiply**. They never add. At least one question on this paper is built on that.
- Percentage change is always measured against the **original** value, not the new one.
- Check every reverse percentage by working forwards again. It takes ten seconds and catches almost every slip.

SECTION A — TECHNIQUE

- 1 Work out $3\frac{1}{4} \div 1\frac{1}{2}$, giving your answer as a mixed number in its simplest form. [3]

Answer

- 2 A number increased by 15% gives 966. Find the number. [2]

Answer

- 3 A restaurant bill of ₹2400 includes a 20% service charge. Calculate the cost of the food alone, before the service charge was added. [2]

Answer ₹

- 4 Write $\frac{7}{40}$ as a decimal and as a percentage, and explain how you could have known in advance that this decimal terminates. [3]

SECTION B — PROBLEMS

- 5 A town's population is 12 000. In the first year it falls by 15%. In the second year it rises by 20%. [4]

- (a) Find the population at the end of the first year. [1]

- (b) Find the population at the end of the second year. [1]

- (c) Find the overall percentage change over the two years, and state whether it is an increase or a decrease. [2]

- 6** Three students take the same test, marked differently. [4]

Meera 45 out of 60 Ravi 72% Sana $\frac{7}{10}$

- (a)** Convert all three scores to percentages. [2]

- (b)** Rank the students from best to worst. [1]

- (c)** How many percentage points separate the best from the worst? [1]

- 7** A tank is $\frac{2}{5}$ full. When a further 30 litres of water is added, the tank is $\frac{7}{10}$ full. [4]
Calculate the total capacity of the tank.

Answer litres

- 8** A shop marks up the cost price of every item by 60%. In a sale, it then takes 25% off the marked price. [4]

(a) Find the single multiplier that takes the cost price to the sale price. [2]

(b) Hence state the percentage profit the shop still makes in the sale. [2]

- 9** Anil spends $\frac{1}{3}$ of his money on books. He then spends 40% of what is left on food. He has ₹360 remaining. [4]

(a) What fraction of his original money did he spend on food? [2]

(b) How much money did Anil start with? [2]
